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CAMELS, PERFUMES, AND PROPHETS

Almost invariably, artists and illustrators portray camels in profile. . . . Seen from the front, a camel's nose is a bulbous, rubbery snout with its upper lip sliding forward beneath it, pouching over its teeth, bulging above its shorter lower lip in such a way that as I squint, I no longer see an animal that looks anything like the way I expect a camel in profile to look full face. What I see is some other creature, something like a sea serpent or a dog-faced dinosaur.—Leila Hadley[1]

The technique of desert raiding should be studied by all travelers. An approaching party may be friend, but is always assumed to be foe. . . . Raiding parties are of two kinds, that whose tribe and yours have no blood-feud, and that where a blood-feud exists. Both want your camels and arms, the second your life as well.—Bertram Thomas[2]

If we are to believe recent geological and paleontological research, the dinosaurs met a sudden, dark, and frigid end when an enormous asteroid hit the Gulf of Mexico about sixty-five million years ago and triggered an ice age. Our warm-blooded mammalian ancestors, better adapted to the cold, enjoyed a resurgence. Around forty million years ago, one of these, the rabbit-size *Protylopus,* arose in North America. At the beginning of the Pleistocene epoch, about three million years ago, the Isthmus of Panama formed, allowing *Protylopus* to migrate to South America, where its descendants, the llama, alpaca, guanaco, and vicuña, thrive in the Andes. In North America, perhaps five hundred thousand years ago, *Protylopus* also gave rise to the modern camel.

The Pleistocene, which ended just ten thousand years ago, was marked by periods of intermittent but vast glaciation. During these frozen interludes the accumulation of ice in the earth's expanding polar caps caused the sea level to drop by as much as several hundred feet, more than enough to expose the bottom of the Bering Strait, at present

less than two hundred feet deep in places. This land bridge, Beringia, allowed the movement of plant and animal species between the eastern and western hemispheres.

During these exchanges in the late Pleistocene, two momentous migrations would occur: human beings moved east from Siberia into the New World, and the camel and horse crossed over in the opposite direction to Asia and onward into Africa. Both these hooved species soon disappeared from North America—perhaps falling prey to large saber-toothed cats, perhaps because of change in their forage caused by climatic oscillation, or perhaps because of the depredations of prehistoric man. Although the horse would be spectacularly reintroduced to the Americas by the Spanish conquistadors, the camel never regained its original birthplace.

Neither did the camel initially thrive in its new home in the Old World. Unlike the fleet horse, the defenseless camel has a top speed of only about twenty miles per hour—an easy meal for the lion or for any other large, swift predator. In Asia's driest regions, particularly Arabia, the camel developed its signature evolutionary advantage: the ability to store and preserve water, enabling it to exist for long periods in the desert far from oases, where large carnivores gathered.

Camels do not store water in their humps, as is commonly supposed, but rather distribute it uniformly throughout their bodies. They are easily able to go days and, in exceptional circumstances, weeks, without water by drinking huge amounts—up to fifty gallons at a time. They conserve fluid through the remarkable ability of their kidneys to efficiently concentrate urine. The first Asian camels were two-humped (Bactrian), but in the hotter Arabian and African deserts, the species evolved toward its familiar single-hump (dromedary) configuration, which reduced its surface area, thus decreasing water evaporation. The dromedary also evolved another water-conserving mechanism, the ability (unusual for a mammal) to passively raise its body temperature up to six degrees Fahrenheit during the heat of the day, minimizing water loss from sweating. To this day, dromedaries predominate in Arabia and Africa, while the Bactrian dwells in Asia.[3]

Initially, both varieties fought a losing battle and were saved from extinction only by the fortuitous appearance of humans. The camel is one of the few animals that can be domesticated. To be bred by humans, a species must simultaneously possess several relatively unusual characteristics:

the provision of appealing and nutritious food, ease of herding, docility, lack of fear of people, resistance to human diseases, and, most critical of all, the ability to breed in captivity. Only a few animals qualify on all counts. Goats and sheep were the first livestock to be domesticated, around ten thousand years ago, followed by chickens, pigs, cattle, and finally, camels. (The donkey, horse, and dog were domesticated primarily for their transport, hunting, and military usefulness, but often wound up in the food chain as well.)[4]

We have little idea of how the most common crop and animal species were first domesticated, and the camel is no exception. On the basis of anthropological evidence, it seems likely that humans began drinking camel milk about five thousand years ago in the Horn of Africa, or perhaps just across the Red Sea in southern Arabia. To this day, Somalis refuse to ride camels, believing that the large, slow, ungainly beasts make their riders easy targets. Today, this region is home to the world's largest population of camels, which are still kept only for their milk. Gradually, mankind discovered other uses for the animal: meat and leather from the males, hair from both genders, and, last but not least, transport.

Until about 1500 BC, the donkey had been the pack animal of choice. Thereafter, nomadic tribes bred camels for transport in huge numbers. If the donkey was a family sedan, able to convey light loads over smooth hard surfaces, then the camel was a Land Rover, whose huge padded hooves allowed it to carry roughly twice as much and twice as fast over long stretches of trackless waste. This capability revolutionized trade over the sands of the Middle East and through the steppes of Asia.[5]

A single camel driver, conducting three to six animals, can transport one to two tons of cargo between twenty and sixty miles in one day. When Tiglath-pileser III of the Assyrians defeated the Arabian queen Samsi about 730 BC, his booty included twenty thousand cattle, five thousand bundles of spices, and thirty thousand camels.[6]

A trader cannot simply sling heavily laden bags across a camel's back. The animal's soft, nonsupportive hump and swaying motion require a frame-and-mattress saddle that distributes the weight of the cargo over its back. Between about 1300 BC and 100 BC, pre-Islamic Arabian nomads refined saddles to the point where they allowed the average pack camel to carry more than five hundred pounds, and in excess of one thousand pounds for the strongest animals. The ultimate configuration, the

north Arabian saddle, has been in continuous use in the Middle East for the past two thousand years.

The Bactrian camels of central Asia are just as highly specialized and carefully bred as those of the Arabian desert, having been domesticated for transport at more or less the same time, around 2500–2000 BC. The slightly cooler and wetter climates of the Asian steppes, Iran, and India favor two-humped design. But whereas the desert Arabs valued the dromedary not only for its transport ability but also for its milk, meat, and hair, central Asians did not. In that part of the world, settled agriculture had already established itself and had spread widely. Central Asians found sheep's wool superior to camel hair, and cow's milk and meat more plentiful and better-tasting. Further, the ox and water buffalo gave the camel a run for its money over short distances, particularly in wet climates, where camels do not thrive.

Thus, as the ancient era wore on, both the size and the range of the more highly valued dromedary population increased and began to impinge on the domain of the Bactrians: first in Syria and Iraq, then in Iran, then in India, and finally in central Asia itself. When the two populations came into contact, the laws of hybridization worked their typical magic. The two types are similar enough to interbreed, and the first-generation offspring of a Bactrian and dromedary (the so-called F1 hybrid) is, as so often happens, a beast possessed of remarkable stamina and strength, perfectly suited to the long distances of the central Asian overland trade. All along the Silk Road, demand mushroomed for these crossbred "super camels," capable of carrying up to half a ton of cargo from China to the western fringes of Asia.

Such beasts of burden can be bred with a Bactrian stallion servicing a large population of female dromedaries, or the other way around. The pairing of a Bactrian stallion and a dromedary mare, however, is used almost exclusively, since one Bactrian stallion can service a large number of dromedary mares, the latter being far more common, even in central Asia. (A similar situation occurs with the powerful all-purpose western pack animal, the usually sterile mule, which is the offspring of a female horse and a male donkey, but for a different reason. The "reversed" F1 hybrid between a stallion and female donkey—the hinny—is rarely bred, because of the difficulties of delivering the large offspring through the birth canal of the smaller donkey mare.)

The inexorable logic of animal husbandry also demands that hardy first-generation crossbreeds *not* mate, as their second-generation offspring are most often small and degenerate; the Arabic and Turkish words for this second-generation crossbreed both translate as "runt." Consequently, the dromedary and the crossbreed predominate in almost all of Africa and Asia; only in the highest and coldest mountains of central Asia, where even the hardy crossbreed cannot thrive, are pure Bactrians bred in any number.[7]

The animal's continued widespread use from Morocco to India to western China attests to its extraordinary transport efficiency. In the modern era, the availability of well-paved roads allows for the even more efficient camel-and-wagon arrangement. The UN Food and Agricultural Organization estimates the present-day world camel population at just shy of twenty million (including 650,000 feral beasts in the Australian outback, whose services became redundant with the advent of the railroad).[8]

Although an exceptional animal and driver might cover as much as sixty miles per day, a more typical day's span is approximately thirty miles. Given the "safe" three-day water capacity of the camel, oases and caravanserais needed to be spaced about one hundred miles apart; this greatly restricted possible routings, particularly in central Asia. Further, since camels cannot negotiate steep, narrow grades, donkeys were required through the mountain passes of the Asian route.[9]

We've already encountered one commodity, silk, that could be shipped great distances in camel packs. But for thousands of years before silk traveled from China to Rome on camels, as well as in ships, another precious cargo found its way thousands of miles from the vast Arabian desert to the ancient world's great centers of civilization in the Fertile Crescent.

The trademark, and the curse, of the Arabian Peninsula is its hot, dry climate. Scarcely a permanent creek penetrates its desert vastness. There are only their ghosts, the parched, meandering wadis (equivalent to the arroyos of the American Southwest), often missed by even experienced travelers until these dormant streams rage as torrents during storms separated by decades.

One part of the peninsula, however, was known in classical antiquity as Arabia Felix—literally, "happy Arabia." The name referred to the

area's fertility. Located in the peninsula's mountainous southwest, in what is now Yemen, it catches the warm, wet winds of the summer monsoon and receives an average of ten inches of rain annually. The southwestern port city of Aden derives its name from the Arabic word for Eden, which accurately describes this rare patch of moist climate. (The rest of the otherwise arid peninsula was known as Arabia Deserta.)

Incense is the general term for frankincense, myrrh, and rarer exotic aromatics that have grown for millennia in Arabia Felix. The earliest Sabaean and Minaean inhabitants, as well as peoples across Bab el Mandeb in Somalia, pioneered its cultivation and export.

Before the arrival of silk and pepper in the West, incense was the premier luxury product of antiquity. To anyone living in Arabia around 1500 BC, the most obvious use for the newly domesticated camel would have been the transport of incense to consumers in the Fertile Crescent and Mediterranean basin. As far back as 3500 BC, Egyptian and Babylonian aristocrats acquired a taste for these fragrant products. Stone monuments from around 2500 BC celebrate incense-trading voyages to the land of Punt: modern-day Yemen and Somalia. The traders might have sailed the Red Sea down its entire length, but, as we've already seen, shallow waters, pirates, and adverse winds cursed this route. It was safer and more reliable to use the overland path north along the Arabian coast of the Red Sea, then west through the Sinai.

The growing cycle also favored the camel route. Growers harvested the crops mainly in autumn and spring, out of sync with the winter monsoon for sailing to Egypt or the summer monsoon for sailing to India, whereas the camel trains could operate year-round.[10] The rigors of navigation in the Red Sea and the peculiarities of the harvests and monsoons drove the peoples of the peninsula to domesticate the camel specifically to carry incense.

The bulk of the trade consisted of two somewhat different products: frankincense, a gum resin produced from *Boswellia sacra;* and myrrh, a fragrant oil produced by *Commiphora myrrha*. Both plant species are scruffy trees several feet tall that grow mainly at high altitude in southern Arabia and neighboring northern Somalia.

Frankincense and myrrh acquired their status as premier luxury items for reasons both sacred and profane. Although our imagination allows us to conceive the sights and sounds of ancient civilizations, their

smells are well beyond modern comprehension. In cramped cities lacking effective sanitation, the nose discerned location as well as any map: the odor of feces from main sewer lines and slaughterhouses; the scent of urine surrounding government buildings, temples, and theaters; or the particularly offensive olfactory assaults of the tannery, the fishmonger, and the cemetery.

Amid such stenches, and where regular bathing in clean water and regular changes of clothes were reserved for only the wealthiest citizens, few substances were as prized as myrrh oil, easily applied as a body lotion and capable of hiding the rank smells of everyday life. Physicians used myrrh liberally in medicinals, and it was also the ancient world's embalming fluid of choice. In addition, incense was the aroma of eros, as attested in this come-on from a well-known, wicked, and deadly biblical adulteress:

> *I have decked my bed with coverings of tapestry, with carved works, with fine linen of Egypt.*
> *I have perfumed my bed with myrrh, aloes, and cinnamon.*
> *Come, let us take our fill of love until the morning: let us solace ourselves with loves.*
> *For the goodman is not at home, he is gone a long journey:*
> *He hath taken a bag of money with him, and will come home at the day appointed.*[11]

Frankincense, while also fragrant, possessed more mystic qualities. This even-burning gum produced delicate, wispy curling smoke that, as envisioned by the ancient world, slowly rose to the heavens, where its appearance and scent pleased the gods. In China and India, funerary rites required the burning of frankincense. In the tabernacles of the early Jews, its opaque tendrils were said to veil the Almighty's very presence.[12]

Pliny wrote that Alexander the Great was particularly fond of burning large amounts of incense on holy altars. Alexander's tutor Leonides admonished him that "he might worship the gods in that manner when he had conquered the frankincense producing races." According to Pliny, Alexander then subjugated Arabia, whereupon "he sent Leonides a ship with a cargo of frankincense, with a message charging him to worship the gods without any stint."[13]

Pliny also gives us a vivid description of the frankincense trade in Arabia Felix. The trees secrete a slick, foamy liquid which collects under the bark. Growers incised the tree, and the fluid spurted onto the ground or onto palm mats, where it dried and thickened. This was the purest, most desirable frankincense, whereas the residual product, stuck to the tree and contaminated with bark, was of second quality. The integrity of the growers amazed Pliny:

> The forest is divided up into definite portions, and owing to the mutual honesty of the owners is free from trespassing, and though nobody keeps guard over the trees after an incision has been made, nobody steals from his neighbor.[14]

Today, bedouin men and women still harvest trees marked to indicate ownership, in accordance with Pliny's observations. Before the Common Era, the growing of incense was confined to the trees' natural habitat in southwest Arabia, and local peoples tapped them only during the hottest part of the year, in May, before the arrival of the cooling, moist southwest monsoon. After drying for a few weeks, the final product either began its northward journey by camel to markets in the Fertile Crescent and Mediterranean, or else was stored for a few more months until the worst of the monsoon storms had cleared and was then sent by sail east to India. The Greek naturalist Theophrastus described the highly trusting "silent trade" transactions that characterized the initial purchases:

> And that when they have brought it, each man piles up his own contribution of frankincense and the myrrh in like manner, and leaves it on guard; and on the pile he puts a tablet on which is stated the number of measures which it contains, and the price for which each measure should be sold; and that, when the merchants come, they look at the tablets, and whichsoever pleases them, they measure and put down the price on the spot whence they have taken the wares, and then the priest comes and, having taken the third part of the price for the god, leaves the rest where it was, and this remains safe for the owners until they come to claim it.[15]

The freshly dried frankincense, a fragile, gummy substance, was packaged in protective wooden cages; myrrh oil, more prone to evaporation, was transported in animal hides. For thousands of years, these two

precious substances, produced in remote and secretive kingdoms, traversed a complex route from the southwest corner of the Arabian peninsula to their ultimate destinations in Babylon, Athens, and the ancient Egyptian capital of Memphis. The historian Nigel Groom writes, "One can visualize the camels of the ancient caravans with frankincense baskets bulging on either side of their saddles or swinging to the weight of myrrh in more compact, tightly closed goatskins."[16]

The Pax Romana changed this pattern. A significant part of the Roman booty went toward the purchase of incense. The earliest Greeks and Romans had probably propitiated the gods with human sacrifice, but in classical Greece and the early Roman republic, this had been replaced with animal offerings. Alongside the sacrificial altar, standing on a tripod, was the *acera,* in which frankincense was placed.[17] So central was the burning of this aromatic to Roman ritual that it was admitted into the empire duty-free, in contrast to the 25 percent duty on most other imports. (The Arch of Titus in the Forum depicts the emperor carrying a balsam shrub in his triumphant march through the capital after the conquest of Jerusalem in AD 70. This plant yielded one of the most expensive aromatics.)[18] With the prosperity of the empire grew its demand for incense, and the gradual extension of Roman control south toward Arabia Felix during the first and second centuries AD made both the maritime and the camel routes safer and cheaper.

As demand rose, growers added a second and third annual crop, which yielded a lower-quality product than the traditional May harvest. Cultivation spread eastward to Zufar, in modern Oman.

This expansion shift added inhospitable distance to cargoes bound for Rome. Some of the frankincense and myrrh from these new growing areas was shipped directly from the eastern Arabian ports of Qana and Moscha to Berenice on the Red Sea and thence to Alexandria. But most of the incense trade was conducted by camel; the king of Arabia Felix, wishing to control the lucrative market, saw that the bulk of it passed overland through the city of Shabwah, in the east of Arabia Felix.

Pliny described how the incense, after being collected, was carried to Shabwah, where "a single gate is left open for its admission." Failure to use the assigned gate—a sure sign of smuggling—was punishable by death. The land route seems to have been the monopoly of a single tribe,


Incense Lands and Routes
Myrhh
Frankincense
Black Sea
Mediterranean Sea
Gaza
Alexandria
Jerusalem
Beersheba
Petra
Ma'an
EGYPT
Aelana
al-Bad
MESOPOTAMIA
Myos Hormos
Caenopolis
Coptos
Leucos Limen
Berenice
Leuce Come
Yathrib (Medina)
Makkah (Mecca)
Red Sea
Nile R.
Thumala
Tabala
Najran
ARABIA FELIX
Marib
Thomna
Shabwah
Moscha
Syagrus
Cana
Bab el Mandeb
Muza
Ocelis
Dioscordia
Mosyllum
Cape of Spices
Avelites
Malao
Mundus
INDIAN OCEAN
500 MILES

whose priests oversaw the harvesting and transport of the crop. Pliny variously identifies the tribe as the Gebanitae or the Minaeans.

The priests at Shabwah took 10 percent of the cargo as an import tax. The incense then had to travel to Thomna, the capital of the country of the Gebanitae or Minaeans, who controlled the onward trade. Pliny records that the journey from Thomna to Gaza, about 1,500 miles, took sixty-five days, about twenty-three miles per day. Along the way, expenses were incurred. Among others:

> Fixed portions of the frankincense are also given to the priests and the king's secretaries, but beside these the guards and their attendants and the gate-keepers and servants also have their pickings: indeed, all along the route they keep on paying, at one place for water, at another for fodder, or the charges for lodging at the halts.[19]

Each camel's load to Rome incurred a total purchase and transport cost of approximately a thousand denarii, or about two denarii per pound. The highest-grade frankincense—so judged by its whiteness, brittleness, and ease of burning—sold in Rome for six denarii per pound. The lowest grade sold for three denarii per pound, about the same price as black pepper. (The denarius, a small silver coin weighing one-eighth of an ounce, roughly corresponded to a day's wage of a skilled laborer. Thus, a pound of frankincense represented roughly a week's worth of wages for a skilled laborer; a pound of myrrh, about two weeks' worth.) By comparison, the most precious aromatics, such as balsam from Palestine, sold for up to 1,000 denarii per pound.

What frankincense lacked in price, it more than made up for in quantity. Alone among the aromatics, frankincense was shipped in camel-load (about 500 pounds) quantities and was therefore by far the most important trade commodity of the era. If we take Pliny's accounting at face value and estimate the total cost of transporting a camel load of incense from Arabia Felix to Rome at approximately 1,000 denarii, then, at an average retail price of five denarii per pound, a total profit of 1,500 denarii per camel load was possible.

Incense generated prosperity all along its supply chain. This wealth was distributed among middlemen, those who serviced the caravans, and the camel drivers themselves, each capable of guiding up to six animals at a time. Snaking their way slowly along the western (Red Sea) coast of

the peninsula, these caravans connected the incense sources of Arabia Felix with well-off consumers in the Fertile Crescent, and later, in Greece, Rome, and Byzantium. All along this route, great marketplaces prospered, particularly at the Sabaean-Gebanite cities of Shabwah, Thomna, and Marib. Another group, nomadic tribesmen, also prospered by plundering the rich incense supply chain. The product that arrived from eastern Arabia via Gaza and Alexandria on the wharves of Puteoli had traveled as much as four thousand miles.

The confusion over exactly who controlled the trade is in large part due to the difficulties of doing research in modern-day Yemen and Saudi Arabia. For most of the twentieth century, the key ancient Minaean-Gebanite town of Marib was off-limits to Westerners. When in 1951 the imam of Yemen finally granted permission to the renowned American archaeologist Frank Albright to visit Marib to help solve this mystery, his party was immediately driven off at gunpoint by unhappy locals. Archaeologists have found fragmentary but tantalizing Minaean inscriptions as far away as Memphis in Egypt and Delos in Greece, suggesting the presence of Arabian trade diasporas thousands of miles from their homeland.

As the domestication of camels for transport spread north and east, other legendary inland hubs, such as Palmyra, Samarkand, and Shiraz (in modern-day Syria, Uzbekistan, and Iran, respectively), swarmed with camel traders, caravan drivers, and the merchants of many nations. Each city became, in its turn, wealthy and powerful. Today, perhaps the most visible remnants of the incense trade are the magnificent stone temples and tombs at Petra, the capital of Nabataea, in what is now southern Jordan.

This mysterious kingdom of sun worshippers flourished between 300 BC and the fall of Rome, and their prosperity hinged on their control of the northern third of the Arabian incense track. Likewise, the Mediterranean terminus of the camel route at Gaza also thrived on the trade. The incense sent by Alexander to Leonides—fifteen tons of frankincense, three tons of myrrh—came from Gazan warehouses plundered by Alexander on his way from Tyre to Egypt in 332 BC. By that date, Gaza was already a very old and wealthy place, sitting on a large funeral mound and having been besieged several times in the preceding centuries by the Assyrians.

By the time the incense reached Egypt, the easygoing honesty of Arabia Felix had completely disappeared. Again, Pliny:

> At Alexandria, on the other hand, where the frankincense is worked up for sale, by Hercules, no vigilance is sufficient to guard the factories! A seal is put upon the workmen's aprons, they have to wear a mask or a net with a close mesh on their heads, and before they are allowed to leave the premises they have to take off all their clothes.[20]

The ancient incense trade was thus no different from the modern cocaine and heroin trades: relatively safe around the raw agricultural source, but highly risky around the finished product and its ultimate consumers.

The effects of incense in its ultimate destination, Rome, were less salutary. Along with silk, aromatic imports drained the empire of silver; Nigel Groom estimates that about fifteen million denarii per year were spent on the ten thousand camel loads of incense bound for the capital. As long as plunder from abroad arrived on the wharves, there was no problem; Seneca's fortune alone was reputed to be almost one hundred million denarii. But in the second century, as the conquests ceased and Romans became ever more extravagant, those more poetically than economically inclined may be forgiven for concluding that the power of the empire evaporated in a haze of incense.[21]

Although frankincense and myrrh spread prosperity to cities and towns all along the caravan route, one place among them—a small western Arabian oasis situated about halfway between Yemen's producers of incense and its consumers in the faraway eastern Mediterranean and Fertile Crescent—would mesmerize the civilized world. There, the incense trade catalyzed the birth of Islam, whose military, spiritual, and commercial impacts transformed medieval Asia, Europe, and Africa. Riding on a rising tide of global trade along the land and sea routes of Asia, Islam came to dominate that continent's spiritual as well as its commercial life.

The saga of the new religion begins with the ancestors of the desert Arabs, who were sedentary farmers working marginal oasis plots. About 3,000 to 3,500 years ago, they first mastered camel domestication, which gave them the ability to challenge the stark, terrifying Arabian wilderness. Even with this newfound mobility, their situation was precarious, huddling in oases during the deadly, rainless summers and foraging the desert fringes with goats and camels the rest of the year.

This grim nomadic existence afforded them a precious geographic remoteness that spared them conquest. The two great western predator states of the post-Roman era, the Byzantine and the Persian Sassanid empires, both sought to regain the past glories of Trajan and Darius from each other: Byzantium endeavored to grab Mesopotamia from Persia, while Persia in turn hoped to regain Syria and Egypt from Byzantium. Locked in a continuous life-and-death struggle, these powers devoted no significant attention to the exotic, impoverished desert dwellers to the south. There was one exception to Arab remoteness and independence, and that was the monsoon-swept, fertile, incense-producing Arabia Felix, which became a luckless pawn in this ancient version of the Great Game.

The harsh and lawless environment of the desert shaped both economic and religious life on the Arabian Peninsula and has to this day left its mark on the culture of the Muslim world. Survival in Arabia, with its lack of a central authority, was, and remains, utterly dependent on the good efforts of the family and the tribe.

Western notions of individual autonomy and rule of law simply do not apply in the desert. An attack on one tribesman is an attack on all, and in a landscape where a murderer can quickly and quietly slip away, it matters little whether the accused is guilty or innocent. His entire clan is held accountable for *thar*—retribution. The resulting skein of honor and revenge, so familiar in the modern Middle East, is eternal, seemingly without beginning and without end. When the first recourse of victims is to their cousins, and not to the police or to an independent judicial system, poverty and political instability are the usual outcomes.

In such a barren and impoverished landscape, a major source of sustenance is often theft from the tents and caravans of neighboring tribes. The signal military operation of the desert is the *ghazu*—a raid mounted on horses (which are both faster and easier to control at a full gallop than the camel). The raiders swiftly and deftly execute these attacks so as to avoid causing casualties that would trigger *thar*.[22] Recall the trilemma of trade: whether to trade, protect, or raid. In the absence of any authority beyond the tribe, entrepreneurs will invariably choose to raid.

The pre-Islamic desert dwellers prayed to many gods, and Islam appropriated many early Arab religious beliefs and practices. The early Arabs erected shrines to numerous deities; the holiest was Kaaba at Mecca,

a large granite block in one corner of which was embedded black stone, probably of meteoric origin. It is not certain if the Kaaba was dedicated to the main Arab deity, al-Llah, or to one of the lesser Arab deities, Hubal. The people of the ancient Middle East commonly worshipped meteoric fragments; the silk-loving Roman emperor Elagabalus, described in the Introduction, was a Syrian who began his career as the high priest of a temple housing such a celestial remnant in the town of Emesa (modern-day Homs in Syria). When he became emperor, he dismayed the Romans by bringing the rock along with him and building another temple for it in the capital.[23] (Twenty-two years after the death of Elagabalus, the leadership of that most ecumenical of empires would fall to Emperor Philip, an Arab.)

By AD 500, the desert Arabs had come into frequent contact with Christians and Jews. The Jews likely had migrated south after the conquest of Jerusalem by Nebuchadnezzar in 586 BC and established palm plantations in the Hejaz. Similarly, Christianity diffused into Arabia both from Byzantium in the north and from the south across Bab el Mandeb in Coptic Christian Abyssinia. Christians and Jews often taunted the Arabs about their polytheistic beliefs and their lack of an overarching creed and an afterlife. Consequently, a sense of religious inferiority arose among the desert dwellers, along with a pent-up desire for a comprehensive belief system of their own.

Exactly how Mecca became a bustling commercial center is something of a mystery; it produced nothing of value, was not a great center of consumption or of government, and had little strategic worth. Some historians have suggested that its main advantage was its position: Mecca lay approximately midway on the two-month journey up the Arabian Peninsula, so far from both Byzantium in the north and Abyssinian-run Yemen in the south that it was relatively safe from the depredations of both. This cannot be the major reason for its prominence, however. The role of the incense trade in the city's rise is also uncertain: there is controversy as to whether or not the main caravan route bypassed the town (in contrast to Medina, through which the incense highway almost certainly ran).[24] Mecca sits in a dry and barren valley, and during the pre-Islamic period was utterly dependent for its food on the gardens and farmlands of Taif, seventy-five miles away.[25] In a narrow sense, Mecca may be thought of as a miniature, parched, landlocked Arabian version of Venice, whose food

supply and rhythms of daily life hummed to the tunes of trade, whether or not it actually sat on the main incense route.

The real reason for the early prominence of Mecca in pre-Islamic Arabia may lie in the Kaaba stone and the several nearby shrines to the other desert gods. Each year, the faithful made the pilgrimage, known as the hajj (which was only much later adopted by Islam), to venerate and circumambulate the Kaaba stone and black stone. The hajj in no small part contributed to Mecca's wealth and power.

By the late fifth century AD, the Quraish tribe, led by a sheikh named Qussay, moved in from the north, took over Mecca, and then fended off invasions by both the Byzantines and Abyssinnians. Qussay next convinced the Quraish and the surrounding tribes that it was more profitable to trade and to protect the caravans than to raid them. Taxing traders and selling them safe conduct, it turned out, paid better than plundering a shrunken, fearful traffic.[26] The Quraish continued to settle in Mecca in increasing numbers, grew wealthy, and gradually drifted away from their intensely communal, nomadic heritage. Their lives henceforth revolved around trade, not the precarious existence of oases and desert tents.

Beginning around AD 500, Abyssinia converted to Christianity and became a regional power closely associated with its co-religionists in Byzantium. The last ruler of an independent Arabia Felix, the handsome Yusuf Asai (also known as Dhu Nuwas and "the man with the hanging locks"), converted to Judaism in the early sixth century and proceeded to slaughter and enslave thousands of Christians in his kingdom. In AD 525, in response to the anti-Christian atrocities of Yusuf Asai, the Abyssinians attacked across Bab el Mandeb and overwhelmed his army. The despondent king was said to have ridden into the sea on horseback.[27]

The defeat of the Yemenite Jewish monarch and the resultant domination of Arabia Felix by Abyssinian Christians set in motion a chain of events whose effects reverberate to the present day. In AD 570, an Abyssinian proconsul in Arabia Felix named Abraha rebelled against his king and established a rival empire on the peninsula. A staunch Christian, with an army backed by African elephants transported from across Bab el Mandeb, Abraha was goaded by the Byzantine emperor Justinian into attacking Mecca, by that time the last pagan holdout in Arabia. The unfortunate elephants, however, although fearsome weapons on most of the ancient world's battlefields, were not well-suited to the searing sands of

Arabia and succumbed, either to disease or to the harsh climate, just outside the gates of the city. The Meccans had never seen such creatures and, being unfamiliar with the basics of animal ecology and microbiology, credited divine intervention. The year AD 571 became known in Arabia as the "Year of the Elephant."[28] That same year saw the birth of the Prophet Muhammad into a lesser branch of the Quraish, and his arrival was imbued forever after by Muslims with the mythic elephantine event. He became, of course, a trader.

Had Abraha and his pachyderm allies succeeded at Mecca, Muhammad, had he been born at all, might have wound up a Christian monk. The historical Muhammad is at best an indistinct figure; the first written accounts of his life did not appear for more than a century following his death, and even these were distorted by the ideological needs of his early chroniclers. Certain basic facts, however, seem beyond dispute. Orphaned at an early age, he was raised by an uncle, Abu Talib, a prosperous trader. Although Muhammad probably spent his formative years observing and participating in his uncle's business, there is no direct record of his early professional endeavors. What is more certain is that at around age twenty-five, he found himself in the employ of an older widow, Khadija, who also ran a prosperous trading enterprise. We do not know precisely what goods her caravans carried, but dates, raisins, and leather from nearby Taif; frankincense from Yemen; and textiles from Egypt and beyond would certainly have been among them.

As a woman, she did not travel with these cargoes, and so Muhammad rapidly gained experience as her agent in Syria. Impressed with the young man's competence and charmed by his personality, she proposed marriage, and he accepted; Muhammad was now a man with standing and resources.

In his travels, Muhammad encountered Jews and Christians—the "people of the Book"—and felt the power of their seductive belief systems. The fact that both Judaism and Christianity were associated with hated foreign powers limited their appeal and drove Muhammad and his Arab countrymen to seek their own path. The Arabs' sense of longing was amplified by their revulsion at the materialistic ways of Mecca's newly wealthy Quraish commercial aristocracy, perceived as having turned their back on the ancient tribal codes of behavior.[29] In the words of Islam's great Western historian, Maxime Rodinson:

> The traditional virtues of the sons of the desert were no longer the sure road to success. Greed, and an eye to the main chance, were much more useful. The rich became proud and overbearing, glorying in their success as a personal thing—no longer a matter for the whole tribe. The ties of blood grew weaker.[30]

By the late sixth century, then, many Arabs were moved by twin needs: to create a single unifying identity in opposition to the two foreign-derived monotheistic religions, and to develop a political force to counter the wealth and corruption of the Quraish. In this turbulent socioeconomic atmosphere, al-Llah, who emerged alone from the gods of the desert, forcibly dictated through the angel Gabriel's voice the first verses of the Koran to an agonized Muhammad on Mount Hira, just outside Mecca, in 610. The dry tinder of religious fervor was now lit, and almost immediately flared up into the conflagration of conversion and conquest that would engulf much of Asia, Africa, and Europe.

Muslims have long recognized that Khadija's support was critical to the Prophet's ultimate mission: a common Arab saying has it that "Islam did not rise except through Ali's sword and Khadija's wealth." (Ali, Muhammad's cousin and son-in-law, eventually became the Prophet's fourth successor; his murder would split the Muslim world into a Shiite minority and a Sunni majority, who believe and disbelieve, respectively, that the leadership should pass directly through Muhammad's lineage by way of Ali.)

Alone among the world's religions, Islam was founded by a trader. (Muhammad's immediate successor, the cloth merchant Abu Bakr, was also a trader.) This extraordinary fact suffuses the soul of this faith and guides the historical events that ricocheted over the land routes of Asia and the sea-lanes of the Indian Ocean through the next nine centuries. Its traces are visible in today's world, from the modern colonies of Muslim Indians in East Africa to the Lebanese merchants still active in West Africa to the "Syrians" who populated the third-world outposts of Graham Greene's novels.

The most sacred texts of Islam resonate with the importance of commerce, as in this famous passage from the Koran: "O you who believe! Do not devour your property among yourselves falsely, except that it be trading by your mutual consent."[31] The most important passages on trade and commerce, however, are found in the hadith, the collected stories of

Muhammad's life, which offer advice on the conduct of trade from the general: "There is no harm for you [to trade] in the Hajj season," to the specific:

> The buyer and the seller have the option to cancel or to confirm the deal, as long as they have not parted or till they part, and if they spoke the truth and told each other the defects of the things, then blessings would be in their deal, and if they hid something and told lies, the blessing of the deal would be lost.[32]

One narrator, Jabir bin Abdullah, tells of a personal encounter with Muhammad, who offers to buy his troublesome camel, and for which one gold piece is paid. In a demonstration of charity, Muhammad later returns the camel and allows Jabir to keep the gold, informing posterity that at that point in his life, the Prophet might have been taken out of the trade, the trade had not yet been taken out of the Prophet.[33]

Within a very few decades, the new creed would sweep from Mecca to Medina and back, next across the Middle East, and then west to Spain and east to India. In a commercial sense, early Islam can be thought of as a rapidly inflating bubble of commerce; outside lay unbelievers, and inside lay a swiftly growing theological and institutional unity. A detailed recounting of Islam's astonishing initial spread is beyond the scope of this book, but it's worth noting that its lighting speed was due in no small part to the conflict between the new creed, which forbade stealing from fellow believers, but not from infidels, and the economic imperative of the *ghazu* (raid). The Prophet may have been born a trader, but he died a raider. Soon after he was expelled from Mecca in 622, he began attacking that city's infidel caravans. The new religion dictated that all the property of conquered nonbelievers was forfeit, with one-fifth earmarked for Allah and the *umma*—the people—and the rest divided between the victorious troops and their leaders.[34] If a people converted peacefully, their property was spared. Thus, as more distant tribes converted, it became necessary to raid ever farther afield to obtain sustenance from resistant nonbelieving tribes. After the Prophet's death in 632, this process accelerated as some peoples were conquered while others, driven both by the political, spiritual, and military power of the new creed and by a desire to retain their assets, saw the light and converted. Both mechanisms—conquest and peaceful conversion—rapidly drove the

boundaries of Islam farther and farther from their starting point deep in the peninsula.

The Arab armies were stopped six years later at the gates of Constantinople by a string of unusual circumstances, that included the recent ascent of Emperor Leo the Isaurian, a master strategist, and unusually cold winter weather that proved deadly to troops bred in the Arabian climate and to an army supplied by camel. In the words of the Islamic scholar J. J. Saunders, "Had [Constantinople] fallen, the Balkan Peninsula would have been overrun, the Arabs would have sailed up the Danube into the heart of Europe, and Christianity might have lingered, an obscure cult, in the forests of Germany."[35]

The Arabs' first order of business was to feed the newly converted and hungry masses of the peninsula. Since time immemorial, Egypt had been the granary of the Mediterranean, and the Muslim conquest opened wide this supply to the demanding markets of Arabia. The caliph first sent grain by caravan down the incense route, but the new Islamic empire soon began clearing the old sea-level canal between the Nile and Red Sea to create a cheap maritime route between Arabia and its Egyptian food source. As in modern times, strategic considerations dictated the fate of this ancient version of the Suez Canal. Initially, the leaders considered extending it directly to the Mediterranean. This would have made its route nearly identical to that of the present-day canal, but Caliph Omar (the Prophet's second successor, after Abu Bakr) decided against the project, fearing that the Byzantines would use a connection between the Mediterranean Sea and the Red Sea to interfere with the hajj. What can feed can also starve: the grain now flowing down the Red Sea had previously sailed north to Constantinople. The loss of this vital food store in no small part contributed to the decline of Byzantium. A century later, Caliph Abu Jaffar shut down the canal for the last time in order to cut off provisions to Arabian rebels.

Next, Muslim forces wrested control of the eastern Mediterranean from the Byzantines at the battle of Dhat al-Sawari—the Battle of the Masts—in 655. At this early date, the Arabs had yet to assemble a working navy and so manned their vessels with experienced Christian Coptic seamen who, ironically, despising their Greek overlords, abetted one of Islam's greatest victories. At a stroke, the sea routes between the West and India, and China as well, were severed, and would remain so until

Vasco da Gama became the first European to break through into the Indian Ocean eight and a half centuries later.[36]

After their victory at the Battle of the Masts, Muslim navies gradually extended their control over the Mediterranean. In 711 a freed Berber slave, Tariq ibn Ziyad, successfully commanded a daring raid at a rocky promontory in southern Spain, then under the rule of the Goths. The Umayyad invaders celebrated this famous victory, which preceded the Muslim conquest of all Spain by just three years, by naming the rock Jabil Tariq: the mountain of Tariq, or, as it later became pronounced, Gibraltar.

Of the strategic islands in the Mediterranean Sea, Cyprus fell nearly in the first flush of Arab conquest in 649. Crete fell in 827, Malta in 870, and, after more than a century of conflict, the sea's greatest prize, Sicily, in 965. As the new millennium dawned, it must have seemed to the forces of Christendom that what the Romans once called *mare nostrum*—our sea—now swarmed with Muslim ships. So wide was the domain of Muslim conquest and trade in Europe that large numbers of Islamic coins dating to the ninth and tenth centuries have been found as far away as central Europe, Scandinavia (particularly on the island of Gotland, off eastern Sweden), England, and Iceland.[37]

The early Umayyad and Abbasid empires ruled, respectively, before and after 750. They controlled an area greater than that ruled by the Romans and as the sheer extent of their conquests dried up the supply of potential booty, commerce increasingly drove their military priorities. Poor and backward western Europe did not interest them as much as central Asia, with its rich silk route. The Umayyads did not return to Gaul after their defeat at the French city of Poitiers in 732, nor did they react vigorously to the *reconquista* of Spain and Portugal, which began in 718 and culminated in the expulsion of the last Moors (and of the Jews) in 1492.

Muslim armies, by contrast, attacked the far reaches of central Asia again and again, not succeeding until they defeated the Tang Chinese at Talas (in present day Kazakhstan) in 751 and delivered that part of the world, with its profitable caravan routes, into Muslim hands, where it remains to this day. Dramatic conquests often lead to startling serendipities: the most momentous Muslim acquisition at Talas was not territory or silk, but a commodity at once prosaic and precious. Among the

Chinese prisoners taken at Talas were papermakers, who soon spread their wondrous craft into the Islamic world, and then to Europe, changing forever human culture and the course of history.

The early Muslim conquests essentially recreated the Pax Romana, but on an even grander scale. The Umayyad and Abbasid empires were in effect large free-trade areas in which old borders and barriers had been swept away, especially along the Euphrates River, since remotest antiquity the traditional frontier between the East and West. No longer were the three great routes to Asia—the Red Sea, the Persian Gulf, and the Silk Road—competing alternatives; rather, they were an integrated global logistic system available to all parties who recognized the suzerainty of the caliphate.

For nearly the next millennium, Muslim seafaring ran well ahead of Muslim conquest and conversion. Astoundingly, by the mid-eighth century, barely a hundred years after the death of the Prophet, thousands of Muslim (likely Persian) traders had arrived not only in Chinese coastal ports, but in Chinese inland cities as well.[38] By contrast, the first large Chinese oceangoing junks did not venture into the Indian Ocean until about AD 1000, and the legendary eunuch admiral Zheng He would not sail his massive fleet to Sri Lanka and Zanzibar for another four hundred years after that.

Arabic was the lingua franca of the new empire, and Muslim navies patrolled ports and sea-lanes from Gibraltar to Sri Lanka. By the ninth century, Islamic rulers in central Asia had established contact with Volga Khazars, and through them, the Scandinavians; in the East, Muslim contact with China grew brisk via both the Silk Road and maritime routes, and North African traders sent caravans south across the Sahara. Within a few centuries of the Prophet's death, his followers had knitted almost the entirety of the known world into a vast emporium in which African gold, ivory, and ostrich feathers could be exchanged for Scandinavian furs, Baltic amber, Chinese silks, Indian pepper, and Persian metal crafts.[39] Further, the Arabs, invigorated by their conquests, experienced a cultural renaissance that extended to many fields; the era's greatest literature, art, mathematics, and astronomy was found not in Rome, Constantinople, or Paris, but rather in Damascus, Baghdad, and Cordova.

The Pax Islamica was not an unmixed blessing; the border between Occident and Orient shifted west into the Mediterranean, through which free passage was lost to Muslim and Christians alike. In the words of the

historian George Hourani, "Instead of a highway, the Mediterranean became a frontier, a sea of war—a change which ruined Alexandria."[40] Although the Muslim commercial web possessed many advanced features, including bills of exchange, sophisticated lending institutions, and futures markets, no Islamic state ever established the bedrock financial institution of the modern world: a national or central bank.[41]

But this was beside the point. For several centuries after the fall of Rome, the fragments of the old empire suffered in obscurity as backwaters of world commerce, largely ignorant of the commercial and technological revolutions taking place in the Middle East, India, and especially China. Even so, Mediterranean shipping did benefit from the introduction of the Arab triangular lateen sail, which enabled vessels to tack into the wind, a feat not possible with the square rigging of Western antiquity.

This Pax Islamica went completely unchallenged until the eleventh century, when a resurgent Christendom recovered substantial territory in Spain, Sicily, and Malta. Emboldened by these gains, in 1095 Pope Urban II convened the Council of Clermont and called forth the First Crusade, which temporarily regained the Holy Land.

In the twelfth century Saladin followed his conquest of the Fatimids with the ejection of the crusaders from Jerusalem (although he was more than happy to trade with his Christian enemies) and consolidated Muslim power in the Middle East. With Saladin's victories came Islam's high point. Thereafter came a devastating series of misfortunes: the Mongol invasions in the thirteenth century, the plague in the fourteenth, and Vasco da Gama's penetrations of the Indian Ocean in the fifteenth and sixteenth.

In spite of Islam's long decline, Muslim traders dominated long-distance commerce until the sixteenth century, and in many areas, well into the early modern period.